Finite character duality behind the Tate pairing

Result. For every finite field K of size q, finite abelian group G killed by q−1 and automorphism σ, Lean constructs both evaluation isomorphisms between invariant K*-valued characters and the actual quotient Q=G/range(σ−1). Evaluation formulas, both trivial radicals and equal cardinalities are proved. No primitive root, basis, rank, supplied duality isomorphism or nondegeneracy premise. This is the finite character/coinvariant algebraic core; actual Galois H1 and canonical divisor pairings remain separate. [pcc:unverified]

Finite algebraic core	G killed by q−1; Q = G / range(σ−1) Invariant chars ≃ Hom(Q,K*); Q ≃ Hom(chars,K*)
Actual F3 positive	Nonsingular $y^2=x^3-x$; P=(0,0) has exact order 2 Actual two-torsion 4; characters 4; coinvariants 4; units 2
Actual char2 control	Ordinary $y^2+xy=x^3+1$; identity action Characters 1 / coinvariants 2; source annihilation fails

Construction and credit. The actual finite field unit group supplies a primitive(q−1)th root. Source annihilation derives enough roots for the quotient exponent. The invariant-character subgroup descends to all quotient characters by explicit inverse lift/pullback maps. Credited Mathlib finite abelian biduality then constructs the reverse evaluation isomorphism, with its formula proved. The maps, source quotient and character groups are actual homomorphism/quotient types. [pcc:unverified]

Actual positive control. Every finite odd field's actual rational two-torsion is killed by q−1, so the result applies to any automorphism of that actual group. On F3, $y^2=x^3-x$ has nonzero discriminant and actual point(0,0) of exact order2. Its actual rational two-torsion has four elements by the actual cubic-root equivalence. With identity action, invariant characters and coinvariants each have four elements, units have two, and evaluation is perfect. No assumed group table or pairing matrix. [pcc:unverified]

Actual failed-hypothesis control. Over every finite characteristic-two field, the credited $y^2+xy=x^3+1$ model supplies an order-two point. Identity coinvariants have cardinality2 but invariant characters cardinality1. A constructed nonidentity quotient class is killed by every character. Lean proves source(q−1)annihilation fails and directly negates the universal right-nondegeneracy statement after deleting it. The qualified result is retained. [pcc:unverified]

Primary review. Peter Bruin, JTNB23(2011),323–328, DOI10.5802/jtnb.764: all six article pages/proofs/references read; all seven PDF pages including cover fully rendered and inspected. Lemma2.1's finite/annihilation conditions are distinguished from section3's separability/Lang/Cartier steps and Theorem4.1's divisor comparison. Those bridges are not assumed or proved here. The original FR1994 remains unread after separate author403/503 and archive503 routes; Bruin is a separate primary source. Cited books/papers are not automatically marked read. [pcc:unverified]

Validation and remainder. Wholegate34/0failures/0trusted native exceptions; all79 build/processing/provenance files losslessly retained in32objects, five gate files and recursive source/config pins. Lean4.35.0/Mathlib56a00585…, standard foundations/default limits. Exact local PCC replay differs from signed standing. KB/Conductor unavailable; bounded fallback recorded. Continuous H1/Lang/isogeny divisibility/Cartier/polarisation/canonical Weil-Tate-MOV-FR and other fourteen-item gaps remain open. No novelty/corpus promotion, holes/custom axioms/native evaluation/raised limits/new global flags/waivers/native experiment/paid compute/upload. [pcc:unverified]